

CHENNAI INTL, India

WMO: 432790

Lat: 12.994N

Lon: 80.181E

Elev: 16

StdP: 101.14

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.0	20.9	14.9	10.6	28.9	16.1	11.5	27.9	7.9	28.4	7.0	28.5	0.4	320	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.2	39.0	26.0	37.9	26.2	36.8	26.0	28.5	33.3	28.1	32.5	27.7	32.1	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.1	30.7	27.1	22.8	30.6	26.8	22.4	30.4	92.7	33.6	90.5	32.7	89.0	31.9	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.0	6.2	DB	18.3	41.7	1.2	1.4	17.5	42.7	16.8	43.6	16.1	44.3	15.3	45.4	(4)
(5)			WB	16.5	29.7	1.2	0.7	15.6	30.2	14.9	30.6	14.2	31.0	13.3	31.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	25.4	26.6	28.7	31.0	32.8	32.2	30.9	30.2	29.8	28.4	26.7	25.6	(6)	
	DBStd	2.84	1.04	1.27	1.46	1.50	1.92	1.84	1.66	1.44	1.30	1.68	1.28	1.14	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6949	478	465	581	630	707	665	649	626	594	571	502	483	(10)	
	CDD18.3	3908	219	232	323	380	449	415	391	367	344	312	252	224	(11)	
	CDH23.3	46467	1647	2152	3729	5074	6355	5857	5291	4723	4288	3458	2231	1662	(12)	
	CDH26.7	22557	428	757	1664	2724	3900	3496	2883	2358	2021	1347	609	370	(13)	
(14) Wind	WSAvg	2.8	2.1	2.3	2.7	3.3	3.7	3.9	3.6	3.3	2.8	2.0	1.9	2.2	(14)	
(15) Precipitation	PrecAvg	1380	23	5	8	14	30	78	116	149	142	285	361	160	(15)	
	PrecMax	2214	170	50	175	114	254	512	218	395	343	800	1071	691	(16)	
	PrecMin	708	0	0	0	0	0	7	19	25	21	2	12	0	(17)	
	PrecStd	301	38	10	24	25	43	72	54	81	75	159	236	149	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	33.9	36.1	38.4	41.3	40.1	38.1	37.1	36.2	35.1	32.9	31.1	(19)	
		MCWB	23.2	23.8	24.8	27.0	26.0	26.1	25.4	25.9	26.0	25.6	25.3	24.4	(20)	
	2%	DB	30.2	32.2	34.9	36.8	39.8	38.4	37.0	35.9	35.0	34.0	31.2	30.1	(21)	
		MCWB	23.1	23.5	25.0	26.9	26.0	25.7	25.4	25.7	25.8	25.5	24.8	23.8	(22)	
	5%	DB	29.8	31.2	33.8	35.8	38.2	37.2	35.9	34.9	34.1	33.0	30.8	29.2	(23)	
		MCWB	22.9	23.4	25.1	26.8	26.6	25.6	25.4	25.7	25.7	25.6	24.7	23.3	(24)	
	10%	DB	29.0	30.5	32.8	34.7	36.9	36.1	34.9	33.9	33.2	31.9	29.9	28.8	(25)	
		MCWB	22.4	23.2	24.9	26.6	26.6	25.5	25.3	25.7	25.7	25.6	24.5	23.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	26.2	27.6	29.0	29.3	28.6	27.8	28.1	28.1	27.8	27.1	26.2	(27)	
		MCDB	28.7	29.9	31.8	35.3	34.6	33.9	31.7	32.2	31.8	30.9	29.8	28.5	(28)	
	2%	WB	24.7	25.6	26.9	28.2	28.7	27.9	27.2	27.4	27.4	27.3	26.5	25.6	(29)	
		MCDB	28.1	29.4	30.9	32.6	33.5	32.6	31.6	31.7	31.2	30.5	29.1	27.9	(30)	
	5%	WB	24.2	25.0	26.5	27.7	28.2	27.3	26.7	26.9	27.0	26.8	26.1	25.1	(31)	
		MCDB	27.5	29.1	30.6	31.9	33.0	32.4	31.5	31.3	30.8	30.1	28.6	27.5	(32)	
	10%	WB	23.6	24.5	26.1	27.4	27.9	26.9	26.4	26.5	26.6	26.5	25.7	24.6	(33)	
		MCDB	26.9	28.5	30.2	31.7	32.7	32.2	31.4	31.0	30.5	29.7	28.1	27.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	8.9	8.2	9.2	8.8	8.1	7.9	7.7	6.7	6.1	6.7	(35)	
		MCDBR	8.5	9.4	9.7	9.2	10.8	9.8	9.3	9.0	8.7	8.2	7.4	7.4	(36)	
	5% WB	MCWBR	3.0	3.1	3.1	2.8	3.5	3.3	3.2	3.1	3.0	2.9	2.8	2.9	(37)	
		MCWBR	7.3	8.0	8.6	8.4	9.3	9.1	8.4	8.1	7.9	6.9	5.8	5.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.468	0.458	0.504	0.522	0.601	0.538	0.535	0.524	0.519	0.533	0.511	0.516	(40)	
		taud	2.160	2.195	2.074	2.073	1.884	2.077	2.087	2.125	2.139	2.088	2.112	2.050	(41)	
	Ebn at Noon		830	857	824	803	730	768	773	791	800	780	784	772	(42)	
		Edh at Noon	148	148	171	170	202	164	164	160	158	163	154	162	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.97	5.86	6.33	6.49	5.99	5.35	5.01	5.08	5.19	4.64	4.20	4.20	(44)	
	RadStd		0.16	0.19	0.22	0.22	0.30	0.37	0.38	0.27	0.32	0.29	0.40	0.36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page